Digitizing High Risk Vehicle Theft Areas: A Pre-Pandemic & Post-Pandemic Comparison

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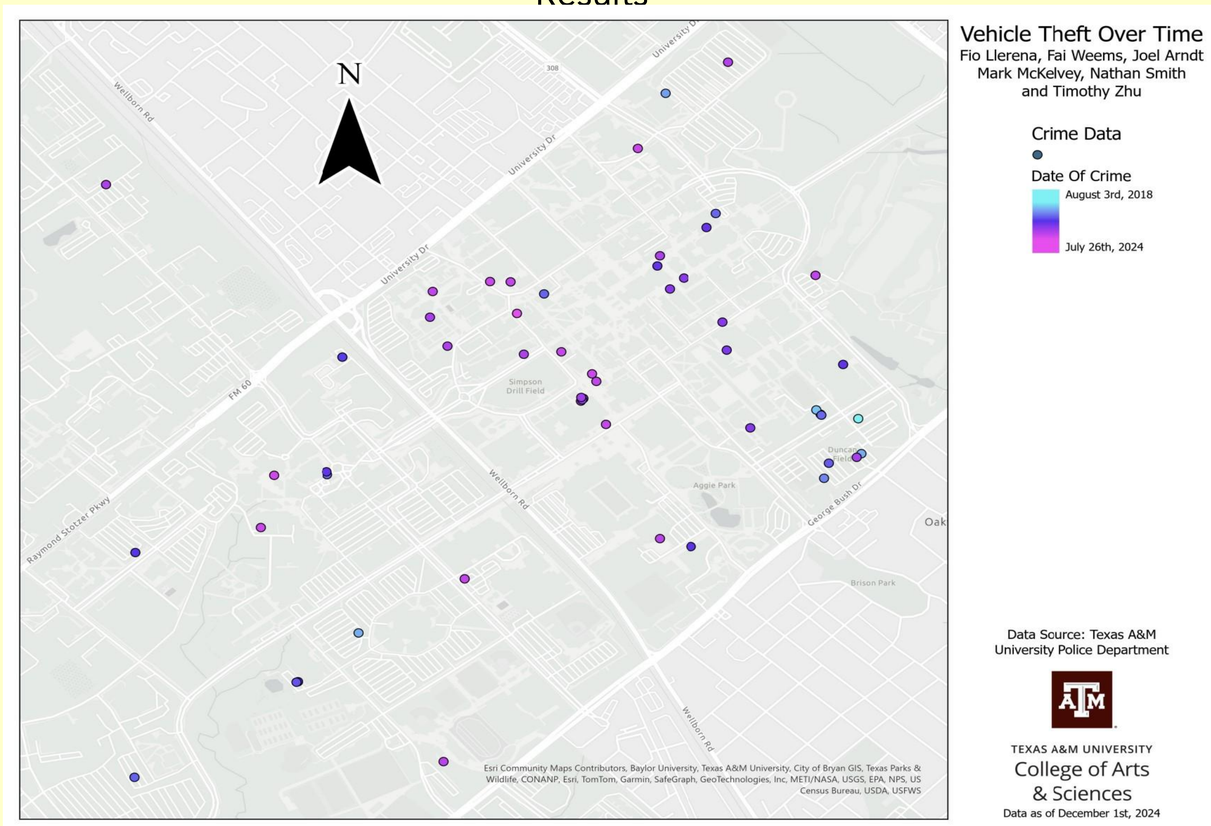
Abstract

As crime data is publicly available through the Clery Act, our group decided to organize this data in order to discern what trends exist among the vehicle theft crimes committed on campus. More specifically, we decided to see whether the pandemic has affected the trends in vehicle crime rates. To accomplish this, we took published data points and plotted them on the ArcGIS Field Maps app, organizing them on ArcGIS Online. We concluded that vehicle theft has increased steadily since the pandemic began, and the highest number of crimes were committed after the pandemic.

Introduction

An NCIB report in 2020 stated that vehicle crime rates increased by 9.2% from 2019 to 2020 (NCIB, 2020). Possible factors for this spike in vehicle theft could include the pandemic's negative impact on mental and physical health, as well as the economic impact (Dureva, 2024). For the TAMU campus, the Clery Act of 1990 requires all crime data to be publicly available. This project seeks to compile this data and compare how vehicle theft rates have changed before, during, and after the 2020 Pandemic. These years are 2018-2019 (Pre-Pandemic), 2020-2021 (Pandemic), and 2022-2024 (Post-Pandemic).

Results



Research Question

How has the pandemic affected trends in vehicle theft crime rates on the Texas A&M campus?

References

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Methods

Data points were taken from the Texas A&M University Police Department website, which reports crimes occurring on and around campus as required by the Clery Act. Reported crimes involving the theft or misuse of a motor vehicle on campus were recorded and classified by date of occurrence and location. Location was then noted to promote a more equal distribution of point collection among researching members; half of the team was assigned to the main TAMU campus, and the other half to the western half of campus. These points were collected in the field and uploaded via the ArcGIS Field Maps app. Lastly, the ArcGIS Online service was then used to compare data by date, as shown in the map above.

Conclusions- Discussion

Various conclusions can be drawn from the results of this experiment. Based on the research question, the data compiled was organized by the date the crime took place, visually portrayed via a color ramp, with teal being pre-pandemic, blue being during the pandemic, and magenta being post-pandemic. The pre-pandemic (2018-2019) era had the least vehicle thefts of all, with only five incidents taking place on the TAMU campus. During the pandemic era (2020-2021), the rate skyrocketed, with 22 vehicle theft incidents happening within these two years, over four times the pre pandemic period of equal length. Finally, in the post pandemic period, 26 vehicle thefts took place in the post-pandemic (2022-2024) period, indicating a steady increase of vehicle theft crime.

One notable trend of the vehicle theft crimes is the geographic distribution and spread of the incidents, with a majority of crimes taking place on the east side of campus, considered the main portion of campus, boxed in by University Drive, Texas Avenue, George Bush Drive, and Wellborn Road. Most likely, this is because of higher concentration of students, parking garages and parking lots, and a higher level of traffic within this region. Further studies in crime rates regarding different forms of crime over the years of 2018-2024 will likely aid in forming a definitive answer regarding whether the pandemic increased crime rates on TAMU campus, though vehicle theft rates alone paint a significant picture regarding this.